Problem 1: Let A be the matrix given by

$$\begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$ Then

(i) Find elementary matrices E_1, E_2, E_3 such that

$A = E_1 E_2 E_3$.

(ii) Compute A^{-1}.

Solution:

$$A = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

Let us find E_1', E_2' and E_3' s.t

$$E_3' E_2' E_1' A = I.$$

$$\Rightarrow \quad A = (E_1')^{-1} (E_2')^{-1} (E_3')^{-1}$$

$$= E_1 \; E_2 \; E_3.$$

Let $E_1' = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $\left(\Rightarrow (E_1')^{-1} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \right)$

$$E_1'A = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Let $E_2' = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ $\left(\Rightarrow (E_2')^{-1} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right)$

$$E_2'E_1'A = \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

Let $E_3' = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}$ $\left(\Rightarrow (E_3')^{-1} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \right)$

& $E_3' E_2' E_1' A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I$

Hence $A = (E_1')^{-1} (E_2')^{-1} (E_3')^{-1}$

$$= \underbrace{\begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{E_1} \underbrace{\begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}}_{E_2} \underbrace{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix}}_{E_3}.$$

(ii) $$A^{-1} = E_3' E_2' E_1'$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & -1 \end{pmatrix} \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \boxed{\begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Problem: Let B' and D' be $M_{m\times n}(\mathbb{R})$. Let B and D be the matrices given by

$$B = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & B' & \\ 0 & & & \end{array}\right) ; \quad D = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & D' & \\ 0 & & & \end{array}\right)$$

Prove that if B' can be obtained from D' by an elementary row (column) operation, then B can be obtained from D by an elementary row (column) operation.

(ii) Conclude from the above that if rank $(B) = r$, then rank $(B') = r-1$.

Solution: Suppose E' be the elementary matrix s.t

$$E'D' = B'$$

Let us define $E = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & E' & \\ 0 & & & \end{array}\right) \quad \text{---} \quad (*)$

E is then an $(m+1)\times(m+1)$ matrix

Then

$$ED = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & E' & \\ 0 & & & \end{array}\right)\left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & D' & \\ 0 & & & \end{array}\right)$$

$$= \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & E'D' & \\ 0 & & & \end{array}\right) = \left(\begin{array}{c|ccc} 1 & 0 & \cdots & 0 \\ \hline 0 & & & \\ \vdots & & B' & \\ 0 & & & \end{array}\right) = B.$$

Claim:

$$\rightarrow \begin{pmatrix} 1 & 0 \cdots 0 \\ 0 & \\ \vdots & E' \\ 0 & \end{pmatrix} \begin{pmatrix} 1 & 0 \cdots 0 \\ 0 & \\ \vdots & B' \\ 0 & \end{pmatrix}$$

The (i,j) entry above will be

when $i = 1$. when $j=1$,

then $(1,1)$ entry $= 1$.

when $j \neq 1$, $(1,j)^{th}$ entry $= 0$

$|||^{ly}$ when $j=1$ & $i \neq 1$, $(i,j)^{th}$ entry $= 0$

$$\begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & & & \\ \vdots & & B' & \\ 0 & & & \end{pmatrix}$$

when $i \neq 1, j \neq 1$

$$(i, j) = \sum e'_{i-1\,k} d'_{k\,j-1} = b_{i-1, j-1}.$$

since $E'D' = B'$

$\blacksquare$.

(ii) Let $\operatorname{rank}(B') = k$.

$\Rightarrow \exists$ elementary matrices $E_1', \ldots, E_s'$ and $F_1', \ldots, F_t'$ such that

$$E_1' E_2' \cdots E_s' B' F_1' \cdots F_t' = \left(\begin{array}{c|c} I_k & O_{k \times n-k} \\ \hline O_{m-k \times k} & O_{(m-k)\times(n-k)} \end{array} \right)$$

$\Rightarrow \exists$ $E_1, \ldots, E_s$, $F_1, \ldots, F_t$

$$E_1 \cdots E_s \, B \, F_1, \ldots, F_t = \left(\begin{array}{c|c} 1 & 0 \cdots 0 \\ \hline 0 & \begin{array}{c|c} I_k & 0 \\ \hline 0 & 0 \end{array} \\ \vdots & \\ 0 & \end{array}\right)$$

$$= \left(\begin{array}{c|c} I_{k+1} & 0_{(k+1)\times(n-k)} \\ \hline 0_{(m-k \times k+1)} & 0_{(m-k)\times(n-k)} \end{array}\right)$$

$$\Rightarrow \quad \operatorname{rank}(B) = k+1 = r$$

$$\Rightarrow \quad k = r-1 \qquad \blacksquare$$

<u>Problem 3:</u> Let $M \in M_{n\times n}(\mathbb{R})$ be such that

$$M = \left(\begin{array}{c|c} A_{k\times k} & C_{k\times n-k} \\ \hline O_{n-k\times k} & B_{n-k\times n-k} \end{array}\right)$$

where $O_{n-k\times k}$ is an $(n-k)\times k$ matrix with 0 as its entries. Then

$$\det(M) = \det(A)\det(B).$$

Proof: Let us prove this by induction on k

when $k = 1$

$$M = \left(\begin{array}{c|c} a & a_{12} \cdots a_{1k} \\ \hline 0 & \\ \vdots & B_{(n-1)\times(n-1)} \\ 0 & \end{array}\right)$$

$$\det(M) = a \det(B)$$
$$= \det(A)\det(B).$$

Assume that the result is proved upto $k-1$

Then

$$M = \left(\begin{array}{c|c} \begin{pmatrix} a_{11} & \cdots & a_{1k} \\ \vdots & & \\ a_{k1} & \cdots & a_{kk} \end{pmatrix} & C \\ \hline 0 & B \end{array}\right)$$

$$\det(M) = a_{11} \det\left(\begin{array}{c|c} \breve{A}_{11} & C_{11} \\ \hline 0 & B \end{array}\right) - a_{21} \det\left(\begin{array}{c|c} \breve{A}_{21} & C_{11} \\ \hline 0 & B \end{array}\right)$$

$$+ \cdots + (-1)^{k+1} a_{k1} \det \left(\begin{array}{c|c} \tilde{A}_{k1} & C_{k1} \\ \hline 0 & B \end{array} \right)$$

$$= a_{11} \det(\tilde{A}_{11}) \det(B) - a_{21} \det(\tilde{A}_{21}) \det(B) + \cdots$$

$$\cdots + (-1)^{k+1} a_{k1} \det(\tilde{A}_{k1}) \det(B).$$

$$= \left(a_{11} \det(\tilde{A}_{11}) - a_{21} \det(\tilde{A}_{21}) + \cdots + (-1)^{k-1} a_{k1} \det(\tilde{A}_{k1}) \right) \det(B).$$

$$= \det(A) \det(B) \qquad \blacksquare.$$

<u>Problem 4:</u> Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be given by

$$T(x_1, \dots, x_n) := (x_1 + x_2 + \dots + x_n, x_1 + x_2 + \dots + x_n, \dots, x_1 + x_2 + \dots + x_n).$$

Then find the eigenvalues and the corresponding eigenvectors of T.

<u>Proof:</u> Let λ be an eigenvalue.

i.e $(x_1, \dots, x_n) \in \mathbb{R}^n \setminus \{0\}$ s.t

$$T(x_1, \dots, x_n) = \lambda(x_1, \dots, x_n)$$

$$f(x_1+x_2+\cdots+x_n,\ x_1+x_2+\cdots+x_n,\ \ldots,\ x_1+x_2+\cdots+x_n) = (\lambda x_1,\ \lambda x_2,\ \ldots,\ \lambda x_n)$$

$$\Rightarrow \quad \begin{aligned} x_1+\cdots+x_n &= \lambda x_1 \\ x_1+\cdots+x_n &= \lambda x_2 \\ &\vdots \\ x_1+\cdots+x_n &= \lambda x_n . \end{aligned}$$

$$\Rightarrow \quad \lambda(x_i - x_j) = 0 \quad \forall\ i \neq j \ . \longrightarrow (*)$$

The eqn $(*)$ is satisfied for $\lambda = 0$.

$\Rightarrow \quad T(x_1, \ldots, x_n) = (0, \ldots, 0)$

$\Rightarrow \quad x_1 + x_2 + \cdots + x_n = 0.$

If $(x_1, \ldots, x_n)$ is an eigenvector corresponding to 0, then $x_1 + \cdots + x_n = 0$.

Notice that if $x_1 + \cdots + x_n = 0$ then $T(x_1, \ldots, x_n) = 0$.

Eigenspace of $0 = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_1 + \cdots + x_n = 0\}$.

If $\lambda \neq 0$

$\Rightarrow \quad x_i = x_j \quad \forall \; i, j$

$$\Rightarrow \quad x_1 = x_2 = \cdots = x_n = y$$

$$\Rightarrow \quad T(x_1, \ldots, x_n) = T(y, \ldots, y) = (ny, ny, \ldots, ny)$$

$$= n(y, \ldots, y)$$

$$\Rightarrow \quad \lambda = n.$$

Hence $\lambda = 0$ and $\lambda = n$ are the only eigenvalues of T.

Eigenspace of $\lambda = n$ is the subspace generated by $(1, 1, \ldots, 1)$. ■

Problem: Find all eigenvalues of the linear transformation $T: \mathbb{R}^\infty \to \mathbb{R}^\infty$ given by

$$T(x_1, x_2, \dots) := (x_2, x_3, \dots)$$

Solution: Let λ be an eigenvalue & $(x_1, x_2, \dots)$ be an eigenvector. Then

$$T(x_1, x_2, \dots) = \lambda(x_1, x_2, \dots)$$

$$\Rightarrow (x_2, x_3, \dots) = (\lambda x_1, \lambda x_2, \dots)$$

$$x_2 = \lambda x_1, \quad x_3 = \lambda x_2 = \lambda^2 x_1, \quad \ldots x_k = \lambda^{k-1} x_1.$$

Hence for any $\lambda \in \mathbb{R} \setminus \{0\}$ we have

$$T(x, \lambda x, \lambda^2 x, \ldots\ldots) = \lambda (x, \lambda x, \lambda^2 x, \ldots\ldots)$$

i.e. λ is an eigenvalue of T. ——— $\blacksquare$

Problem: Let $T: M_n(\mathbb{R}) \to M_n(\mathbb{R})$ given by $T(A) = A^t$.

(i) Find eigenvalues of T.

(ii) Describe the eigenvectors corresponding the eigenvalues.

Solution: Let λ be an eigenvalue. Then $\exists$ a non-zero matrix A s.t

$$A^t = T(A) = \lambda A$$

$$(A^t)^t = T(A^t) = T(\lambda A) = \lambda T(A) = \lambda^2 A$$

$$\Rightarrow \quad A = \lambda^2 A$$

$$\text{or} \quad (\lambda^2 - 1) A = 0$$

$$\Rightarrow \quad (\lambda^2 - 1) = 0$$

$$\Rightarrow \quad \lambda = 1 \text{ or } \lambda = -1$$

(ii) If $T(A) = A \Rightarrow A = A^t$

i.e A is a symmetric matrix.

$$T(A) = -A \quad \Rightarrow \quad A^t = -A$$

$$\Rightarrow \quad a_{ij} = -a_{ji}$$